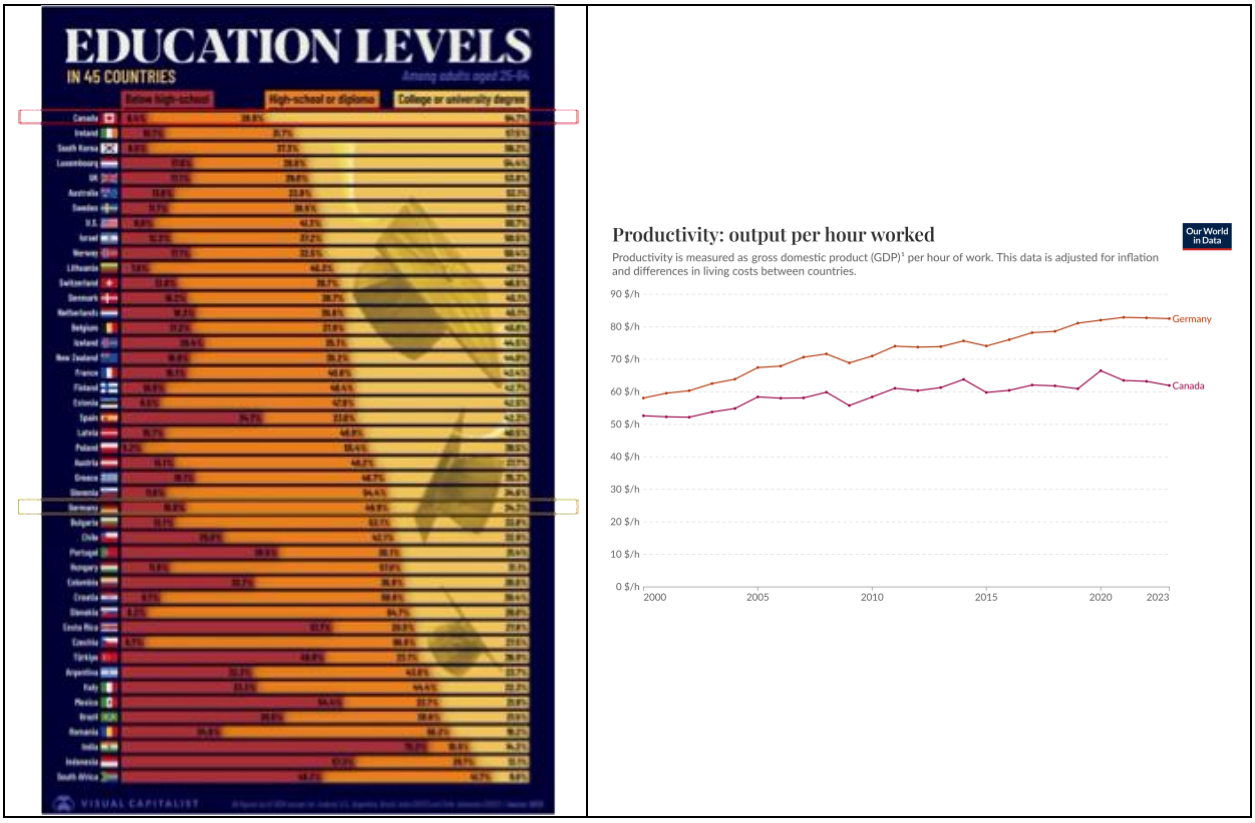
Education and productivity

Consider the two charts provided below. The chart on the left shows education levels across the 45 OECD countries with boxes highlight two countries, Canada and Germany. Canada tops the charts for educational attainment of its citizens, with nearly 2/3 of working age adults holding a university degree. Germany has barely half that number, at 34.3%. The chart on the right shows productivity levels for the same two countries over a nearly 25 year period from 2000 to 2023. In 2023, Germany enjoyed productivity levels that are over 30% higher than Canada's, a gap that has grown substantially over the period highlighted.



1. Juxtaposed, the two charts present a puzzle relative to the way we have discussed productivity in class. Define the puzzle as clearly as you can using concepts from class. For this question, you don't have to explain why the puzzle exists, i.e., what is causing it -- just explain what the puzzle is. Make sure to refer to the data provided and reference both highlighted countries. (Word Count Guidance: 100). 10 marks.

2. Now, focus on Canada. Develop three plausible hypotheses to explain why Canada's strong educational attainment has not translated into faster productivity growth. In your answer, refer explicitly to both charts—education and productivity—and connect your reasoning to relevant growth and productivity concepts from the GMM framework. You should draw on your knowledge of Canada gained from our class discussions. Your answer will be judged on the plausibility, importance and clarity of your hypotheses, as well as the connections you make to the data provided in the two charts. Make sure your three hypotheses are distinct from each other – don't give minor variations of the same hypothesis (Word Count Guidance: ~150 words; 3 x 50). 15 marks = 3 x 5 for each hypothesis.

3. Now, focus on Germany. It is often said—humorously but accurately—that Germany's economy is built around firms that "make the thing that makes the thing that goes in the thing that is the thing." (Think: companies that produce machines that make components—like bearings or sensors—that go into finished products such as vehicles or industrial equipment.) Drawing on this insight, explain how Germany's economic structure might help account for the pattern shown in the two charts. Again, be sure to link the two pieces of data provided in the charts and focus your answer completely on Germany (Word Count Guidance: ~100 words). 10 marks

4. The Government of Canada is preparing its Fall Budget, which—for the first time—will clearly distinguish day-to-day operating spending from long-term capital investment. The new framework aims to prioritize projects that generate lasting economic benefits for Canadians—major resource developments (mines, pipelines), housing, clean-energy infrastructure, transportation networks (railways, ports), and other "generational investments" designed to boost growth and attract domestic and foreign investment.

   Now imagine that Prime Minister Mark Carney visits Ivey the week before the budget is released, and you are granted a five-minute one-on-one meeting with him. He wants your candid view on whether Canada's education system is well aligned to support this new wave of transformative investment—and your ideas for improving it. Carney seeks your opinion on a specific budget proposal he is considering: to cut funding for university research. He tells you:

   > "I'm concerned that our universities aren't delivering clear productivity benefits, so I hesitate to keep pouring money into something that may not be working. Our fiscal outlook is increasingly difficult—deficits are rising, and so is our debt-to-GDP ratio. The proposal on the table is to reduce spending on basic scientific research. My view is this: let other countries do the discovery work—scientific knowledge spreads quickly across borders anyway. Why spend scarce resources on efforts that benefit others as much as us? Given our fiscal constraints and strategic challenges, we should focus on applying new ideas in our industries, not discovering them—be the best at adopting innovations developed elsewhere."

As the starting point for your answer, please state either **"Yes, good idea"** or **"No, bad idea,"** then explain your reasoning. (Word Count Guidance: ~150 words). 15 marks